Data Analysis and Tax Compliance

Statistical reasoning, data and decisions
Block 1: introduction

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What can an audit record tell us?

An administration audits 100 firms and finds discrepancies in 30.

30%

Discussion

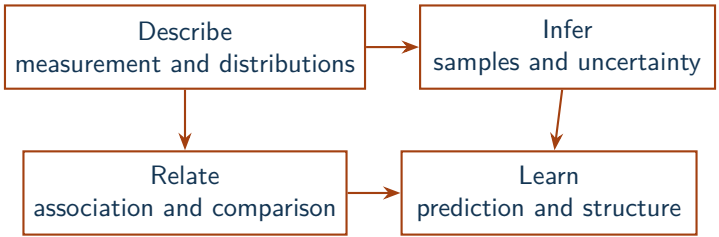
Does this mean that 30% of all firms have a discrepancy?
What would you need to know first?

Hypothetical example. No real tax records or estimates.

The question determines the analysis

Question	What we need
What happened in these records?	Description of the observed data.
What happens in all firms?	Inference, with a defensible sampling or modelling framework.
Which cases need review?	Prediction, evaluated on relevant new cases.
Did a policy change behaviour?	A design that can support a causal comparison.

Our route through the data analysis module



At each stage: what was measured, who is represented, and what does the result justify?

The first meeting

14:30–14:45	Course organisation, survey and opening audit question.
14:45–15:05	A diagnostic discussion of statistical reasoning.
15:05–15:30	A shared dataset: measurement, summaries and interpretation.
15:30–15:40	Break.
15:40–16:00	Fiscal records, tax compliance and human-centred AI.
16:00–16:20	R and the two independent e-tivities.
16:20–16:30	Exit questions and preparation for 4 November.

Our meetings in 2026

Date	Time	Proposed focus
14 October	14:30–16:30	Introduction and launch of independent activities
4 November	14:30–16:30	Debrief and introduction to inference
6 November	10:30–12:30	Bivariate analysis
11 November	14:30–16:30	R: sampling means and bivariate analysis
13 November	10:30–12:30	Introduction to statistical learning
18 November	14:30–16:30	Supervised learning
20 November	10:30–12:30	R laboratory: supervised learning
25 November	14:30–16:30	Unsupervised learning
27 November	10:30–12:30	R laboratory: unsupervised learning
2 December	14:30–16:30	Applications to open data
4 December	10:30–12:30	Applications and synthesis

All meetings: room 424. Dates, times and room from the supplied tentative timetable. Topic allocation is a teaching proposal. The full course timetable also includes Chiara Francioso's meetings.

Theory, e-tivities and R lessons

	E-tivity	Purpose
1	Counting and taxation	From fiscal records to compliance
2	Descriptive statistics	Activities 2A, 2B and 2C
3	Inference	Sampling variability and uncertainty
4	Bivariate analysis	Relationships between variables
5	Supervised learning	Prediction and validation
6	Unsupervised learning	Structure, distances and interpretation

Activities 1 and 2 prepare the return to class on 4 November.
The remaining activities will accompany the later blocks.

R laboratory and statistical learning

1. After bivariate theory: **sampling distributions in R**
   Municipal IRPEF data, repeated samples and the sample mean.
2. **Bivariate analysis in R**
   BES indicators, tables, group comparisons and correlation.
3. **Supervised learning**
   Theory followed by examples in R.
4. **Unsupervised learning**
   Theory followed by examples in R.

Before 4 November

1. Counting and taxation
Allow about 60–90 minutes.
Explore fiscal records and Petty, compare assessments and payments, then connect the evidence to AI.
Bring a short reflection and one question.

2. Descriptive statistics
Allow about 2–2.5 hours, in two sessions.
Review concepts, explore the interactive examples, and analyse the supplied data.
Bring a short analysis and your unresolved questions.

The activities provide formative feedback. Keep your exported answers for the classroom debrief.

Course information and support

- Course announcements and teaching materials: ARIEL course page.
- Contact: silvia.salini@unimi.it.
- Arrange an appointment by email.
- Use the supplied GitHub activity files for practice and save your work locally.
- The official course page will confirm assessment arrangements.

The ARIEL address is retained from the supplied introductory slides.

Class survey and entry test

Data Analysis and Tax Compliance
Academic year 2026/2027

Scan the QR code with your phone.

Complete the class survey and entry test before our diagnostic discussion.



A diagnostic discussion

- A sample contains a million records. Must it represent the population?
- Can two datasets have the same mean and very different distributions?
- A firm has $z = 1.5$. What does this tell us?
- An algorithm flags a case. Is this evidence of an intentional violation?

Discussion

Work in pairs. Explain your reasoning before using a formula.

Our shared dataset

Firm	Sector	Channel	Audited	Turnover (thousand EUR)	Delay (days)
F01	Commerce	Paper	0	18	0
F02	Commerce	Online	0	20	0
F03	Commerce	Online	1	20	0
F24	Manufacturing	Online	1	300	missing

24 fictional firms, one filing per firm for a single reference year.
The file also records whether an audit found a discrepancy.
All records are synthetic. They describe no real firms or tax system.

The unit comes before the variable

Unit of analysis
Here: one firm and its filing for the reference year.
A table of invoices would have a different unit.

Variable
Sector is categorical.
Turnover is quantitative.
Delay is a count of days.
The firm identifier labels the record.

Discussion

If one firm generates 1,000 invoices, how could an invoice-level mean differ from a firm-level mean?

Missing is different from zero

Zero delay
The filing is on time under the definition used in this teaching dataset.

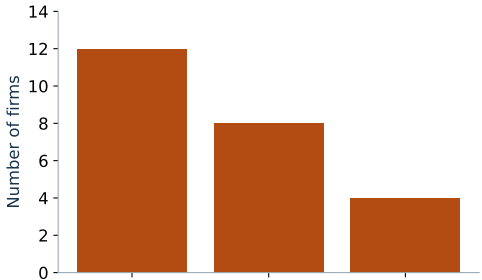
Missing delay
The delay is unrecorded. We do not know whether it is zero.

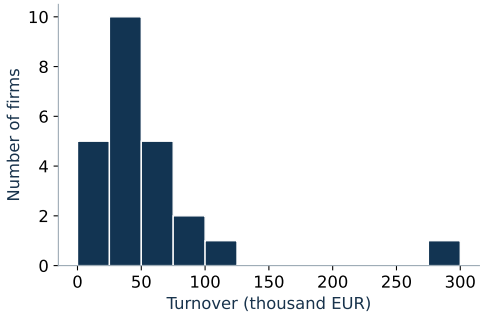
No audit
The discrepancy outcome is unobserved. It cannot be recoded as "no discrepancy".

Discussion

What denominator should appear in a reported percentage?

A graph makes different features visible

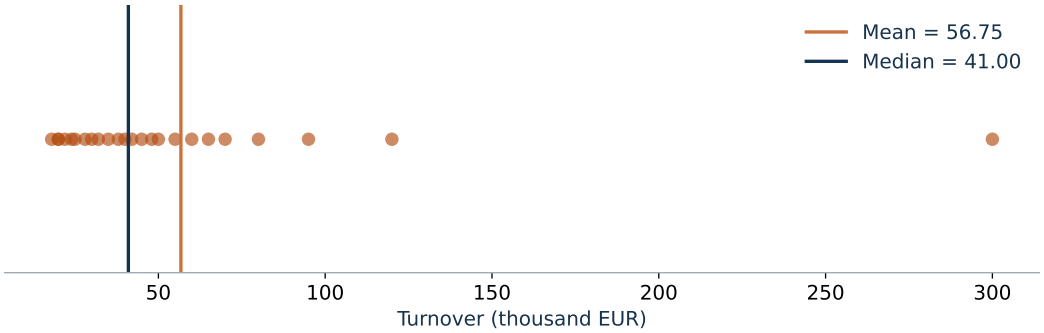




Discussion

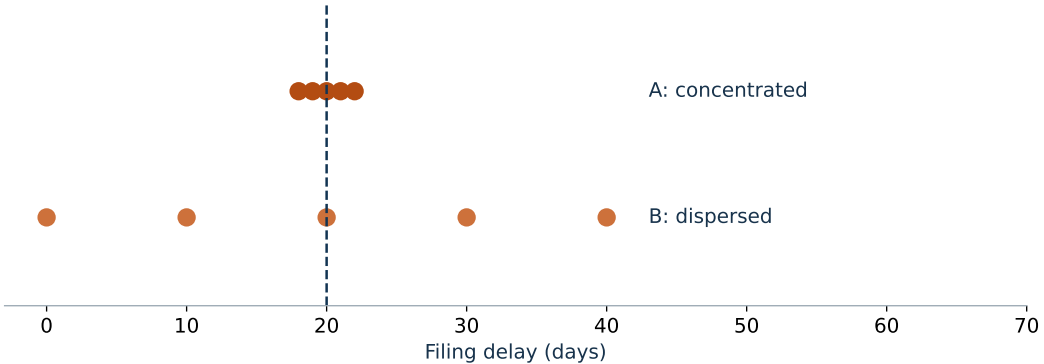
Which graph answers a question about categories? Which reveals the long right tail?

The typical firm depends on the question



Mean: the turnover per firm if the total were redistributed equally.
Median: the middle of the ordered observations.
Both are meaningful. They answer different descriptive questions.

The mean leaves out the spread



Both groups have mean 20 days. Group B varies much more.

A summary of the centre should be accompanied by a description of spread.

Audit outcomes depend on selection

In the teaching data, 10 firms were audited and 6 had a recorded discrepancy.

$$6/10 = 60\%$$

This describes the audited firms. The other 14 outcomes are unobserved.

Discussion

If audits targeted suspected cases, why might this percentage differ from the percentage among all firms?

Synthetic example. A discrepancy is not a legal determination or a measure of intention.

Statistical reasoning begins with a question

1. State a question that can be investigated.
2. Define what the observations and measurements represent.
3. Examine the data and the process that generated them.
4. Choose an analysis appropriate to the question.
5. Interpret the result, its uncertainty and its limits.

Discussion

Where could an error enter before any calculation is made?

Fiscal administration: one root of statistical records

Who is liable? What resources are recorded? What is owed?

Roman censuses registered citizens and property for public administration, including taxation and civic obligations.

Such records made people and resources administratively visible.
Their coverage followed the institutions and purposes that created them.

Discussion

Would a headcount be enough to allocate a tax burden?

ONS, Census-taking in the ancient world. Administrative registers are not yet modern statistical inference.

1086: land, resources and the Domesday survey

- William I's survey recorded landholding, resources and valuations in England.
- Property information helped the Crown understand its fiscal resources.
- It was **not a census of every person**.

Discussion

What does a property register reveal that a headcount does not? Who or what could remain outside its scope?

The National Archives, Domesday Book research guide.

1662: Petty on counting and tax burdens

A Treatise of Taxes and Contributions, chapter VII.

- A headcount can make a fixed charge easy to calculate.
- A title or rank need not measure a person's resources.
- Charging the same amount can impose unequal burdens.

Discussion

Which part is a measurement problem? Which part requires a principle for allocating the burden?

William Petty (1662), A Treatise of Taxes and Contributions, chapter VII, paragraphs 2–6. Paraphrase for discussion.

Same headcount, different possible assessments

	Households	Property value	By headcount	By property
District A	100	800	100	160
District B	100	200	100	40
Total	200	1,000	200	200

Both rules raise 200 monetary units.
Equal household charges imply 12.5% and 50% of property value.
Property-proportional charges imply 20% in each district.

Discussion

Data describe the consequences. What principle would justify the chosen rule?

Invented teaching example, not a reconstruction of historical taxation or current law. Property is the illustrative base.

Assessment and compliance ask different questions

Assessment
Apply a rule to determine what is owed.

With the property rule: A owes 160; B owes 40.

Payment
Compare matched payments with assessed amounts.

A pays 120: 75%.
B pays 36: 90%.

A pays more units; B pays a higher fraction.
Neither ratio reveals unregistered taxpayers or unreported income.

Discussion

What must be defined before calling an indicator a measure of compliance?

Synthetic example. Same obligation, period and reference date. We use modern concepts to interpret history, not to equate different historical institutions.

Statistics also developed beyond the treasury

1662: John Graunt and mortality
Observations on the Bills of Mortality connect recorded deaths and the study of population.

1826: Melchiorre Gioia and statistical knowledge
Filosofia della statistica: a bridge to the wider question of how observations about society inform public decisions.

Fiscal administration is one historical thread. Demography, probability and scientific inquiry also shaped modern statistics.

Graunt (1662), Wellcome Collection; Gioia/Gioja (1826), bibliographic record. Further reading in E-tivity 1.

From fiscal registers to an audit-risk model

Question about a register	Question about training data
Who and what were recorded?	Which taxpayers and outcomes are represented?
Why was the information collected?	How did past audit choices shape the labels?
Which decisions did it support?	What can the prediction justify today?

Discussion

An unobserved audit outcome is not evidence of compliance. A risk score is not a legal finding.

Course synthesis: a contemporary application of the historical questions, not a claim that historical authors anticipated machine learning.

From a predicted risk to a human decision

A model assigns a high risk score to a filing.

Statistical question	Consequence for the decision
Who supplied the training data?	Some firms may be poorly represented.
What does the outcome measure?	Past audit findings may reflect audit selection.
Which errors occur?	False alarms and missed cases have different costs.
How well does it work on new data?	Performance can change across groups and settings.

R makes an analysis reproducible

- R is the statistical computing language used in the module.
- RStudio provides an interface for writing and running R code.
- A script records each step, from reading a file to creating a graph.
- Save the data, code and definitions needed to understand your results.

R Project RStudio Posit Cloud

A short, reproducible analysis

```
d <- read.csv("filings_synthetic.csv")
table(d$sector)
mean(d$turnover_kEUR)
median(d$turnover_kEUR)
sd(d$turnover_kEUR)
sum(is.na(d$filing_delay_days))
```

For each output: name the variable, its unit, and the observations used.
The e-tivity includes a complete script and an explanation of its results.

E-tivity 1: history and a contemporary problem

First: complete Counting, taxation and statistical knowledge and its self-check.

Then: an administration wants to use historical audits to predict future discrepancies.

1. Select one idea from the historical pathway.
2. Explain how it changes your approach to this problem.
3. Identify information you would request before using the model.

Write 250–350 words. Refer to the resource you used.

E-tivity 2: descriptive learning path

Open the descriptive learning path

- 2A: Reading data and graphs.
- 2B: Centre and variability.
- 2C: Heterogeneity and indices.
- A short analysis of the 24 synthetic firms, with documented choices.

Bring back: your answers, a short report, and two questions you would like to discuss.

What you should be able to do on 4 November

- Describe a dataset without confusing the unit, the variable and the population.
- Choose and interpret summaries suited to the data and the question.
- Explain how missingness, selection and extreme values affect an analysis.
- Distinguish a description of observed cases from a claim about a population.

Discussion

Exit ticket: one concept you can explain, one choice you would check, and one question for November.

Reading and resources

For descriptive statistics and inference
Agresti, Franklin and Klingenberg, *Statistics: The Art and Science of Learning from Data*, Pearson.

For statistical learning
James, Witten, Hastie and Tibshirani, *An Introduction to Statistical Learning*.
statlearning.com

For independent work
The two HCAI e-tivity pages, the synthetic dataset, the R script, and the companion Beamer review.